

- 



# COS10025 Technology in an Indigenous Context Project unit

School of Science, Computing, and Engineering  
Technologies

## Seminar Week 4: Indigenous STEM Knowledge

- 
- A 10x10 grid of dots on a white background. The dots are arranged in 10 rows and 10 columns, forming a square pattern. Each dot is a small black circle. The grid is centered on the page.

. . . . .  
. . . . .

## Acknowledgement of Country

We honor and celebrate the rich tapestry of Vietnam's Indigenous peoples, encompassing 54 distinct ethnic groups, including the Kinh, Tay, Thai, Muong, Co Tu, Ede, Khmer, Hmong, Cham and many others, whose ancestral stewardship of this land spans millennia.

We pay our deepest respect to our ancestors past, present and future for preserving Vietnam's cultures, history, timeless traditions and deep spiritual connection to its mountains, rivers, seas and forests.

We recognize their enduring heritage, resilience, and invaluable contributions to the nation's diverse identity, committing to learn from and uplift their stories with reverence and gratitude.



. . The worshipping of Hung Kings is a unique beauty in the spiritual culture of Vietnamese people

. . . . .  
. . . . .



# Indigenous STEM Knowledge

# What will be covered today?

- Indigenous Knowledge of food & medicine
- Indigenous Land management & Contemporary land management
- Indigenous Seasons and Calendars

• • • • • • • • • •  
• • • • • • • • • •

. . . . .  
. . . . .  
. . . . .

# Indigenous Knowledges & Western Knowledges – Food and Medicine

. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .

# Indigenous Knowledges

We must remember, that Indigenous knowledges are specific to specific Indigenous groups across Vietnam are diverse, and have very different cultures, languages and customs. So too, are their health knowledges – particularly those around certain plants and animals which are specific to place.

However, the constant amongst all Indigenous ethics, is the holistic nature of Indigenous health and the inclusion of the “metaphysical, holistic, oral/symbolic, relational and intergenerational” (Levac et al. 2018)

Indigenous health is much broader than western models of health – as it includes interpersonal relationships, emotions, as well as physical and spiritual.

Even Indigenous understandings of learning and intelligence are intrinsically linked to nature and land, and the duality of the lived experiences of people, plants and animals.

. . . . .  
. . . . .

# Western Knowledges

Western notions of intelligence differ strongly from Indigenous understandings – they ignore spirituality and connection, instead focusing on objectivity and “scientific”\* rational.

The western model is much more anthropocentric, positioning humans at the top of the hierarchy of worldly order, and not plants, animals or the land.

However, Levac et al. (2018) states:

“there are schools of Western philosophical thought that lend themselves more to Indigenous ways of knowing, such as feminist and narrative theories that privilege storytelling”

• • • • • • • • • •  
• • • • • • • • • •

# Indigenous ways of knowing and health

- Holistic health
- Traditional healing and medicine
- Caring for Country as necessary for holistic wellbeing
- Sorry business
- Men's and Women's business (gender specific Lore, custom and roles)
- Health being a communal state, rather than individual





# Traditional healing and medicine

## What is traditional healing and medicine?

Traditional healing incorporates many modes, including:

Balancing spiritual and physical wellbeing

Healing songs

Bush rubs and medicines e.g. herbal remedies

Healing foods

Traditions passed down over generations

Health interventions specific to local mobs



# Traditional Healing

Indigenous Vietnamese's traditional healing was our primary form of health care and was conducted by our traditional healers, clevermen or elders.

Today, our people still use traditional healers, alongside western medicine. Our traditional healers have extensive place-specific knowledge on how to treat and diagnose illness in our mobs.

Like all our knowledge, health knowledge is passed generation to generation orally. Some knowledge is sacred and secret, others wildly shared. Some of us are born healers and gifted this knowledge from our ancestors – as you will see in the film next week.

Our knowledges worked well for thousands of years protecting us from illness and disease – however, associated government policies, removal of land and forest, pollutions, introduced foods, lifestyles and diseases, have had devastating affects on mob.

. . . . .  
. . . . .

# Traditional Healing

## Holistic and Diagnostic System

- Combines internal healing with physical techniques: herbal medicine, acupuncture, acupressure, diet.
- Uses the **Four Diagnostic Methods**:
  - **Vọng** – Observation
  - **Vấn** – Listening & Smelling
  - **Vấn** – Inquiry (patient interview)
  - **Thiết** – Palpation (pulse, body)

## Emphasis on Prevention

- Focuses on **balance between body and environment** to prevent illness.
- Promotes long-term health with:
  - Detoxifying herbs
  - Immune-boosting tonics
  - Daily self-care practices (*đưỡng sinh*)
- Example: Herbal remedies help balance **yin–yang** and **qi–blood**.



# Traditional Healing

## Root-Cause Healing

- Targets the **underlying cause**, not just symptoms.
- Techniques include:
  - **Acupuncture**: pain relief, self-healing stimulation (see image)
  - **Massage & acupressure**: improves blood flow, reduces stress
- Supports internal balance and total-body recovery.



## Relevance to Modern Lifestyle

- Addresses modern health challenges: stress, obesity, diabetes, high blood pressure.
- Integrates with wellness habits:
  - **Meditation, qigong, balanced nutrition**
- Enhances both **preventive care** and **quality of life**.

• • • • • • • • • •  
• • • • • • • • • •

Traditional healing is place-specific – as it relates to the weather, landscape, ailments, disease and plants and animals of a particular area.

Traditional healing methods from Northern VN, may not work in the Southern, because they have been accumulated over centuries of trial and error, specific to the place.

This is also important, as it speaks to the health practices of the ecosystem too – remember that Indigenous holistic health includes the health of country, and ecosystems too.

While the learning of traditional healing may come from a traditional healer or the elders, the knowledge is understood by people to include spiritual, ancestral and cosmological influence.

Our healing knowledge's, like all our knowledges are conveyed through story – and they often speak of the knowledges being derived from dreams, visions or from the dreaming itself.

. . . . .  
. . . . .

# Food & Plants as Medicine

Traditional foods, and knowledge of food properties are vital to Indigenous health.

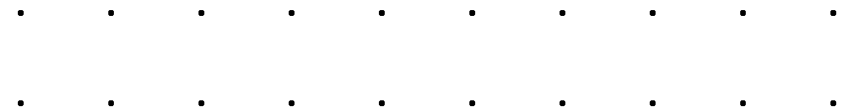
Studies (Milburn 2004) have shown that people can reduce and even reverse some of the negative health problems we have as a result of Western introduced foods, lifestyles and illnesses (such as hypertension, cardiovascular issues and diabetes) by returning to a traditional food diet.

80% of the world's population still relies on traditional medicines to meet their primary health care needs ([WHO, 2008](#)) and nearly 25% of modern medicines are derived from nature, many of which were derived from traditional uses ([De et al., 2012](#))

Plants are very important to Indigenous healing, as they form the base of many of our bush medicines.

Plants were crushed and heated, before being applied to the skin. Or made into saps to be lathered on skin. Or smoked and inhaled. Or boiled and inhaled. Or drunk in tea. Or eaten.

We are going to explore some traditional bush medicines on the next few slides.



# Food & Plants as Medicine

Some results of ethnobotanical study on medicinal plants used by local Van Kieu ethnic people of Bac Huong Hoa Nature Reserve, Vietnam (2019):

Total of **111 species** from **102 genera** and **46 plant families** were recorded.

Dominant families: Euphorbiaceae, Compositae, Leguminosae, Apocynaceae, Rutaceae, Rubiaceae.

**Participants:** 93 informants (aged 20–81), including 5 traditional healers. Most respondents were illiterate (91.4%), reflecting the oral nature of knowledge transmission.

Images (left to right): Cây cam thảo - chè Vằng - đinh lăng - sâm Ngọc Linh





**Table 2**  
List of medicinal plants used by Vao Kieu ethnic people, Vietnam.

| No. | Family        | Scientific name, voucher specimen number                              | Vao Kieu name                    | Uses <sup>1</sup>                                                     | Parts used <sup>2</sup>     | Preparation method                            | IUCN Red list | EVM <sup>3</sup> |
|-----|---------------|-----------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------|-----------------------------|-----------------------------------------------|---------------|------------------|
| 1   | Acanthaceae   | <i>Acanthaceae</i> Vahl KRB 0077801                                   | Da ngyt                          | CA: 2 (Oedema)                                                        | Le (2)                      | Decoction / oral                              | LC            | B                |
| 2   |               | <i>Justicia gendarussa</i> Burm.f. KRB 0077720                        | Thanh tía, P luyt                | MU: 5 (Sprain/strain of joint NDS)                                    | Le (2), Ro (1), Le & Ro (3) | Crushed / fomentation                         |               | A                |
| 3   | Asteraceae    | <i>Pseudanthemum latifolium</i> B. Hansen KRB 0077586                 | Rhau dau khop, Con thi, Xuan ha  | CA: 2 (Elevated blood pressure)                                       | Le                          | Juice / oral                                  | LC            | A                |
| 4   |               | <i>Asteraceae</i> KRB 0077761                                         | Xi ngat, Thạch xương bồ          | RE: 3 (Throat symptom, Cough(2))<br>DI: 2 (Teeth/gum disease)         | Ro                          | Decoction / oral<br>Alcohol maceration / chew |               | B                |
| 5   |               | <i>Celosia argentea</i> L. KRB 0077806                                | Ra tui, Muc gá                   | NE: 2 (Paralysis)                                                     | Le & Ro                     | Decoction / oral                              |               | A                |
| 6   |               | <i>Dyphena amabilis</i> (L.) Morykin & Clematis KRB 0077710           | Sinh, Rau muót, Dầu giáy         | SK: 4 (Ame(2), Urticaria(2))                                          | Le                          | Crushed / apply to skin                       |               | A                |
| 7   | Anacardiaceae | <i>Croton avicariae</i> L. KRB 0077734                                | Ning hoa trắng, Rhau paru li nhỏ | MU: 4 (Musculoskeletal disease)                                       | Le                          | Crushed / fomentation                         | LC            | A                |
| 8   |               | <i>Choerodendron ciliatum</i> (Roxb.) B.L.Burt & A.W.Hill KRB 0077811 | Xoan nhú                         | CA: 2 (Oedema)<br>SK: 2 (Ame)                                         | Le                          | Crushed mix with salt/ apply to skin          |               | A                |
| 9   |               | <i>Morinda indica</i> L. KRB 0077227                                  | Lá xoài                          | PR: 2 (Post-partum symptom)<br>DI: 1 (Abdominal pain)                 | Le<br>Ro                    | Decoction / sauna<br>Decoction / oral         |               | B                |
| 10  | Apiaceae      | <i>Centella asiatica</i> (L.) Urb. KRB 0012513                        | Rau má nhá                       | FE: 3 (Fibrosarcoma uter(2), Gonorrhea female (1))                    | Wp                          | Decoction / oral                              | LC            | A                |
| 11  | Apocynaceae   | <i>Alstonia scholaris</i> (L.) R. Br. KRB 0077620                     | Sa re, Sítá                      | SK: 2 (Ame)                                                           | La                          | Raw / apply to skin                           | LC            | A                |
| 12  |               | <i>Cryptolepis dubia</i> (Burm.f.) M.R.Alsola KRB 0077815             | Cây diu, An lư                   | FE: 1 (Vaginitis)                                                     | Ro                          | Decoction / oral                              |               | B                |
| 13  | Ebenaceae     | <i>Dryas volubilis</i> (L.f.) Benth. ex Hook.f. KRB 0077655           | Rừng Nhút                        | EN: 2 (Diabetes non-insulin dependent)                                | Le & Ro                     | Decoction / oral                              | LC            | B                |
| 14  |               | <i>Ebenaceae</i> (RIDL) Woodson KRB 0077827                           | Chú, Thân lư li quế              | SK: 4 (Scabies)                                                       | Le                          | Juice mix with salt / apply to skin           |               | B                |
| 15  |               | <i>Ebenaceae</i> (Pierre ex Hua) Woodson KRB 0077819                  | Má mít, Thà lư                   | SK: 5 (Ame)                                                           | La                          | Raw / apply to skin                           |               | A                |
| 16  | Araliaceae    | <i>Streptocaulon javanicum</i> (Lour.) Merr. KRB 0077688              | Hà thà A, Hà thà A trắng         | GE: 5 (Bleeding(4), Health maintenance/prevention)                    | La                          | Raw/ apply to injury places (4)               | LC            | A                |
| 17  |               | <i>Tabernaemontana beccari</i> Lour. KRB 0077598                      | Tân tiều, Sừng trâu              | SK: 2 (Ame)                                                           | La                          | Raw / apply to skin                           |               | A                |
| 18  |               | <i>Schefflera heptaphylla</i> (L.) Foidl KRB 0077652                  | Khá lư, Kháng                    | UR: 2 (Urine symptom)<br>FE: 2 (Abdominal pain)                       | Le                          | Juice / oral                                  |               | A                |
| 19  | Asteraceae    | <i>Trema palmeri</i> (Roxb. ex Lindl.) Vis. KRB 0077804               | Sơ prau, Dư đư rừng              | FE: 2 (Post-partum symptom/complaint oth.)                            | Wp                          | Decoction / oral                              | LC            | A                |
| 20  |               | <i>Braconia fragrans</i> (L.) Koe Gowl. KRB 0077629                   | Rhau Pư màu vàng, Thiệt một lư   | MU: 5 (Musculoskeletal disease(3), subluxation (2))<br>CA: 2 (Oedema) | Le                          | Crushed / fomentation                         |               | A                |
| 21  | Bignoniaceae  | <i>Crotonia cajuputi</i> L. KRB 0077659                               | A Lôi, Dư dư                     | GE: 4 (Health maintenance)                                            | St & Le                     | Decoction / oral                              | LC            | B                |
| 22  |               | <i>Oreocarya indicum</i> (L.) Kurz KRB 0077724                        | A Pư                             | DI: 2 (Abdominal pain)<br>GE: 3 (Fever)                               | Ro                          | Decoction / oral                              |               | A                |
| 23  | Caricaceae    | <i>Carica papaya</i> L. KRB 0077786                                   | Dư đư, Má hồng                   | MU: 3 (Low back symptom)<br>DI: 1 (Viral hepatitis)                   | Fr<br>Ro                    | Juice / oral (3)<br>Decoction / oral (1)      | LC            | A                |
| 24  | Combrataceae  | <i>Combretum indicum</i> (L.) DePhilippis                             | Tà tưng, Dư gư                   | BL: 1 (Anaemia other/unspecified)                                     | St & Le                     | Decoction / oral                              |               | B                |
| 25  |               | <i>Combretum pilosum</i> Roxb. ex G.Don KRB 0077817                   | Tuần bư lư, Tà tưng              | DI: 1 (Worms)<br><br>GE: 2 (Bleeding)                                 | Fr<br>Le                    | Raw / oral<br>Crushed/ apply to injury places | LC            | A                |
| 26  | Commelinaceae | <i>Collisia fragrans</i> (Lindl.) Woodson KRB 0077685                 | Rhau, Lưc vàng                   | MU: 2 (Low back symptom, Muscle pain)                                 | Le                          | Crushed / fomentation                         |               | A                |
| 27  | Compositae    | <i>Acmella oleracea</i> (L.) R.K.Jansen KRB 0077772                   | Cúc nút áo                       | DI: 2 (Teeth/gum disease)                                             | Ro                          | Crushed/ chew                                 | LC            | A                |

(continued on next page)



Table 2 (continued)

| No. | Family        | Scientific name, voucher specimen number                                    | Viet Nam name                   | Uses <sup>1</sup>                                                                                                                         | Parts used <sup>2</sup>    | Preparation method                                                                                    | IUCN Red list | BVM <sup>3</sup> |
|-----|---------------|-----------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------|------------------|
| 28  |               | <i>Agrostis conyzoides</i> (L.) L. KRB 0077716                              | Bật khưu, Cứt lợn, Cỏ hôi, Khia | DI: 1 (Abdominal pain)<br>RE: 1 (Stomach cramps)<br>GE: 4 (Bleeding)                                                                      | Le                         | Juice / oral<br>Decoction / sauna<br>Crushed / apply to skin                                          |               | A                |
| 29  |               | <i>Artemisia vulgaris</i> L. KRB 0077738                                    | Cây cứu, Ngải cứu               | DI: 1 (Rectal bleeding)                                                                                                                   | St & Re                    | Decoction / oral                                                                                      |               | A                |
| 30  |               | <i>Bidens pilosa</i> L. KRB 0077644                                         | Sa rum, Răng, Đom đóm           | EN: 2 (Diabetes non insulin dependent)                                                                                                    | Le & Ro                    | Decoction / oral                                                                                      |               | B                |
| 31  |               | <i>Bianca habarifera</i> (L.) DC. KRB 0077657                               | A Fúc, Đại bí                   | MU: 7 (Rheumatoid/4), Fracture: radialis/ulna(1), Arteriosclerosis/complaint(2)<br>FE: 2 (Menstruation irregular, Genital disease female) | Le                         | Crushed / fermentation<br>Decoction/bath                                                              |               | A                |
| 32  |               | <i>Chromolaena odorata</i> (L.) R.M.Hing & H.Rob. KRB 0077657               | A Hoàng, Cỏ lảo                 | DI: 7 (Abdominal pain(5), Diarrhoea)<br>GE: 3 (Bleeding)<br>FE: 3 (Menstruation irregular, Genital disease female(3))                     | Le                         | Juice / oral<br>Crushed / apply to injury places<br>Decoction/bath                                    |               | A                |
| 33  |               | <i>Eclipta prostrata</i> (L.) L. KRB 0012545                                | Cỏ mực                          | GE: 1 (Fever)                                                                                                                             | Wp                         | Decoction / oral                                                                                      | LC            | B                |
| 34  |               | <i>Euphorbia scabra</i> L. KRB 0027238                                      | Cây lết, Cỏ mực                 | SK: 1 (Lump/swelling localized)                                                                                                           | Re                         | Alcohol maceration / apply to skin                                                                    |               | A                |
| 35  |               | <i>Sigesbeckia orientalis</i> L. KRB 0077589                                | S Nam, Giếng, Hy thán           | GE: 1 (Bleeding)                                                                                                                          | Le                         | Crushed / apply to injury places                                                                      |               | A                |
| 36  | Cornaceae     | <i>Cornus ardisioides</i> Jack. KRB 0077523                                 | Đào Rộc                         | GE: 1 (Poisoning by medical agent)                                                                                                        | Le & St & Re               | Decoction / oral                                                                                      |               | B                |
| 37  | Crotonaceae   | <i>Rhyphyllum pinnatum</i> (Lam.) Oen KRB 0077589                           | Thuần hồng, Bì cứt              | GE: 5 (Fever(4), Psychological disorders)                                                                                                 | Le                         | Juice / oral                                                                                          |               | B                |
| 38  | Euphorbiaceae | <i>Acalypha indica</i> L. KRB 0077615                                       | Hoa dĩa đỏ, Tai tượng           | ER: 3 (Eczema/atopic eczema)                                                                                                              | Le                         | Juice / oral                                                                                          |               | B                |
| 39  |               | <i>Alchornea rugosa</i> (Lour.) Mill. Arg. KRB 0077799                      | Đom đóm                         | GE: 3 (Fever)                                                                                                                             | Wp                         | Decoction / Sauna                                                                                     |               | A                |
| 40  |               | <i>Hemaria riparia</i> Lour. KRB 0077662                                    | A Tiếp, Rủ ri                   | DI: 3 (Abdominal pain)                                                                                                                    | Le                         | Juice / oral                                                                                          | LC            | A                |
| 41  |               | <i>Jatropha curcas</i> L. KRB 0077630                                       | Bông, Cây Dầu mè                | MU: 3 (Rheumatoid)<br>SK: 3 (Ache, Burn/scald, skin disease(3))                                                                           | Re (3)<br>Le (1)<br>Le (4) | Decoction / oral (3)<br>Crushed mix with salt / apply to skin (1)<br>Raw / apply to injury places (4) |               | A                |
| 42  |               | <i>Mucoranga clematoma</i> (Blume) Mill. Arg. KRB 0077674                   | Sa lang, Xà lang                | DI: 2 (Worms/other parasites, Diarrhoea)<br>GE: 1 (Poisoning by medical agent)                                                            | Le                         | Juice / oral                                                                                          |               | B                |
| 43  |               | <i>Mallotus barbatus</i> Mill. Arg. KRB 0077794                             | Lục ục, Bùn bọp                 | FR: 1 (Post-partum symptom/complain: etc.)                                                                                                | Wp                         | Decoction / sauna                                                                                     |               | B                |
| 44  |               | <i>Melastoma malindangense</i> (Rafin. ex Blume) Rehb. & Zol. KRB 0077730   | Xà lang pết, Lắc lác lác        | FR: 1 (Sleep disturbance)                                                                                                                 | Le                         | Decoction / oral                                                                                      |               | B                |
| 45  |               | <i>Ricinus communis</i> L. KRB 0077671                                      | A Hồng Thập, Thầu dầu           | DI: 1 (Diarrhoea)                                                                                                                         | Fr                         | Decoction / oral                                                                                      |               | A                |
| 46  |               | <i>Triadica cocklinchensis</i> Loos. KRB 0077792                            | Sỏi                             | FR: 1 (Post-partum symptom/complain: etc.)                                                                                                | St                         | Decoction / Sauna                                                                                     |               | B                |
| 47  |               | <i>Vernicia fordii</i> (Hemsl.) Ait. Shaw KRB 0077661                       | Trầu                            | DI: 1 (Vomiting)                                                                                                                          | Le                         | Cook with pork/oral                                                                                   |               |                  |
| 48  | Gelsemiaceae  | <i>Gelsemium elegans</i> (Gardner & Chapm.) Benth. KRB 0077690              | Lá ngón                         | SK: 3 (Scabies)                                                                                                                           | Le                         | Crushed / apply to skin                                                                               |               | B                |
| 49  | Hypericaceae  | <i>Cratogeomys jamaicensis</i> (Jacq.) Benth. & Hook.f. ex Dyer KRB 0077670 | Lá nghè, Thanh ngạnh            | RE: 1 (Throat symptom/complaint)                                                                                                          | Ba & Ra                    | Decoction / oral                                                                                      | LC            | B                |
| 50  | Isocarpaceae  | <i>Iodex caribaea</i> Turcz. KRB 0077821                                    | Má nang, Mệ thông               | GE: 1 (Malignancy NOS)                                                                                                                    | Re                         | Decoction / oral                                                                                      |               | A                |
| 51  | Lamiaceae     | <i>Callisperm rubella</i> Lindl. KRB 0077593                                | Tím Hoàng                       | DI: 1 (Tooth/gum disease)                                                                                                                 | Le                         | Crushed / chew                                                                                        |               | B                |
| 52  |               | <i>Clerodendrum cynophyllum</i> Turcz. KRB 0077683                          | Đắng cây                        | SK: 1 (Scabies)                                                                                                                           | Le                         | Decoction / bath                                                                                      |               | B                |

(continued on next page)

Table 2 (continued)

| No. | Family      | Scientific name, voucher specimen number                    | Van Kieu name               | Uses <sup>1</sup>                                                                                                                                     | Parts used <sup>2</sup>       | Preparation method                                                                                       | IUCN Red list | DVM <sup>3</sup> |
|-----|-------------|-------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------|---------------|------------------|
| 53  | Lauraceae   | <i>Chondrodendron glandulosum</i> Lindl. KRB 0077740        | Sò Rò                       | GE: 1 (Anemiasis/NOE)                                                                                                                                 | Ra                            | Decoction / oral                                                                                         | EM            | B                |
| 54  |             | <i>Chromolaena balansae</i> Lecomte KRB 0041908             | Cây Re hươg, Pò lai Pò B    | PE: 4 (Post-partum symptom)                                                                                                                           | Le(1), St & Le(3)             | Decoction / Sauna                                                                                        |               | B                |
| 55  |             | <i>Litsea cubeba</i> (Lour.) Pers. KRB 0077505              | Vò to vò, Màng tạng         | PE: 3 (Post-partum symptom)                                                                                                                           | Le(2), St & Le(1)             | Decoction / Sauna                                                                                        |               | A                |
| 56  | Leguminosae | <i>Pitheche xuyana</i> (Meisn.) Hook. f. KRB 0077706        | Hàng se ca, Hàng se         | PE: 3 (Post-partum symptom)                                                                                                                           | St & Le(3)                    | Decoction / Sauna                                                                                        | VU            | B                |
| 57  |             | <i>Dalbergia tonkinensis</i> Frain KRB 0077645              | Huò, Luồng khio, Khio       | SE: 2 (Insect bite/sting)<br>DE: 2 (Teeth/gum disease)                                                                                                | Le<br>Ro & St                 | Crushed / decoction<br>Crushed/ chew                                                                     |               |                  |
| 58  |             | <i>Derris elliptica</i> (Wall.) Benth. KRB 0077742          | Sò to hươg                  | SE: 2 (Skin disease)                                                                                                                                  | Le & St                       | Crushed / decoction                                                                                      |               |                  |
| 59  |             | <i>Strychnine variegata</i> L. KRB 0077741                  | Vàng                        | DE: 2 (Diarrhoea)                                                                                                                                     | Ba                            | Decoction / oral                                                                                         |               |                  |
| 60  |             | <i>Platyrrhiza macrophylla</i> (Willd.) Merr. KRB 0077735   | Tóp mớ, Chè chát            | UR: 3 (Kidney symptom/complaint)                                                                                                                      | Ro & St & Le                  | Decoction / oral                                                                                         | LC            | B                |
| 61  |             | <i>Milletia elliptica</i> (Roth.) Steud KRB 0077813         | Một cốc, Dây một            | MU: 4 (Fracture, Sprain(3))<br>DE: 1 (Abdominal pain)                                                                                                 | Ra<br>Ro,                     | Alcohol maceration / oral<br>Decoction / oral                                                            |               |                  |
| 62  |             | <i>Senna alba</i> (L.) Roxb. KRB 0077728                    | Muồng trâu, Đậu bét         | SE: 3 (Pruritus(4), Itch(1))<br>DE: 1 (Dyspepsia/indigestion)                                                                                         | Le                            | Crushed/ apply to skin<br>Decoction / oral                                                               |               |                  |
| 63  |             | <i>Senna tora</i> (L.) Koch. KRB 0077774                    | Sò Toong, Đậu lều gậy       | RE: 4 (Cough)                                                                                                                                         | Se                            | Juice / oral                                                                                             |               |                  |
| 64  |             | <i>Spacholobus suberectus</i> Dunn KRB 0077703              | Péc, Huyết đằng             | PE: 1 (Post-partum symptom/complaint oth.)                                                                                                            | St                            | Decoction / oral                                                                                         |               | A                |
| 65  |             | <i>Tachyloba triquetra</i> (L.) H. Ohashi KRB 0077687       | Từ Gông                     | RE: 4 (Cough)                                                                                                                                         | Ro,                           | Decoction / oral                                                                                         |               |                  |
| 66  |             | <i>Hibiscus rosa-sinensis</i> L. KRB 0077638                | Dâm bụt                     | UR: 1 (Kidney symptom/complaint)                                                                                                                      | Le                            | Juice / oral                                                                                             |               |                  |
| 67  |             | <i>Sida rhombifolia</i> L. KRB 0077634                      | Mềng xéng rừng, Lá hoa vàng | SE: 2 (Lump/swelling localized, Acne)                                                                                                                 | Ro,                           | Alcohol maceration / apply to skin                                                                       |               |                  |
| 68  |             | <i>Melia azadirachta</i> L. KRB 0077825                     | Xoan                        | SE: 1 (Acne)                                                                                                                                          | Le                            | Juice mix with salt / apply to skin<br>Crushed mix with salt / apply to skin                             |               | A                |
| 69  |             | <i>Parsonia racata</i> Pierre KRB 0077095                   | Hoàng đằng                  | DE: 2 (Diarrhoea, Abdominal pain)                                                                                                                     | Ro                            | Decoction / oral                                                                                         |               |                  |
| 70  |             | <i>Tinospora crispa</i> (L.) Hook. f. & Thomson KRB 0043184 | Dây đau lưng, Sà liêh rô    | DE: 1 (Abdominal pain)                                                                                                                                | St                            | Decoction / oral                                                                                         |               |                  |
| 71  |             | <i>Mimosa pudica</i> L. KRB 0077576                         | Lưu hồ, Ra dầy, Trich sủ    | SE: 3 (Acne(2), Lump/swelling localized)                                                                                                              | Ro                            | Alcohol maceration / apply to skin                                                                       | LC            | B                |
| 72  | Mimosaceae  | <i>Antecarpus heterophyllus</i> Lam. KRB 0077642            | Mít, Ba nai                 | FE: 2 (Fibromyoma uterus, Gonorrhea female)<br>PE: 9 (Post-partum symptom(4), Itch(1), symptom(2))<br>GE: 3 (Poisoning by medical agent(2), Fever(1)) | Wp<br>Le & Ra<br>Le & Ba & Fr | Decoction / oral<br>Decoction/Sauna (4), Decoction/oral (5)<br>Decoction /Sauna (2), Decoction/ oral (1) |               |                  |
| 73  |             | <i>Azadirachta indica</i> L. KRB 0077771                    | Si                          | DE: 2 (Abdominal pain)                                                                                                                                | Le & Ba                       | Decoction/oral (2)                                                                                       |               |                  |
| 74  |             | <i>Azadirachta indica</i> L.f. KRB 0077902                  | Từ nam, Ngái                | DE: 2 (Liver disease NOS)                                                                                                                             | Fr                            | Raw / oral                                                                                               |               |                  |
| 75  |             | <i>Psidium guajava</i> L. KRB 0077721                       | Ôi                          | DE: 1 (Vomiting)                                                                                                                                      | Fr & Le                       | Juice / oral                                                                                             |               |                  |
| 76  |             | <i>Rhodomyrtos tomentosa</i> (Alston) Hassk. KRB 0077677    | K lá, Sín                   | DE: 9 (Abdominal pain)                                                                                                                                | Le                            | Decoction / oral                                                                                         |               | A                |
| 77  |             | <i>Averrhoa carambola</i> L. KRB 0077782                    | Đuối                        | DE: 2 (Abdominal pain)                                                                                                                                | Fr&Le                         | Decoction / oral                                                                                         |               |                  |
| 78  |             | <i>Scheuchzeria palustris</i> L. KRB 0077731                | Nhột, Rón, Lột              | GE: 2 (Fever)<br>SE: 3 (Skin disease)                                                                                                                 | Le<br>Ba                      | Decoction / oral<br>alcohol maceration/ apply to skin                                                    |               |                  |
| 79  |             | <i>Phyllanthus urticaria</i> L. KRB 0069214                 | Chó đẻ                      | DE: 3 (Abdominal pain)<br>CA: 2 (Hemorrhoids)                                                                                                         | Le<br>Wp                      | Juice / oral<br>Decoction / oral                                                                         | LC            | A                |
| 80  | Piperaceae  | <i>Piper corratum</i> Roth. KRB 0077635                     | Lá lốt, Xà lốt              | DE: 5 (Teeth/gum disease(4), Abdominal pain)                                                                                                          | Le & Ro (4)<br>Ro (1)         | Crushed/ chew (4)<br>Decoction / oral (1)                                                                |               |                  |
| 81  |             | <i>Plantago major</i> L. KRB 0077747                        | Bông mã đề, Mã đề           | PS: 2 (Sleep disturbance)<br>UR: 3 (Kidney symptom/complaint)                                                                                         | St & Le & Ro (2),<br>Ro(3)    | Decoction / oral                                                                                         |               |                  |

(continued on next page)

Table 2 (continued)

| No. | Family           | Scientific name, voucher specimen number                                        | Viet. Name                 | Uses <sup>1</sup>                                                                                                                         | Parts used <sup>2</sup>                     | Preparation method                                                            | IUCN Red list | DV/M <sup>3</sup> |
|-----|------------------|---------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------|---------------|-------------------|
| 82  |                  | <i>Scoparia dulcis</i> L. KRIB 0077743                                          | Cơm thảo nam, Cơm thảo đất | GE: 3 (Fever)                                                                                                                             | St & Le                                     | Juice / oral                                                                  |               | A                 |
| 83  | Pontederiaceae   | <i>Cymbopogon flexuosus</i> (Nees ex Steud.) W. Watson KRIB 0015413             | Sả                         | DE: 1 (Liver disease NDS)<br>GE: 1 (Health maintenance)                                                                                   | Wp<br>St                                    | Decoction / oral<br>Decoction / sauna                                         |               | B                 |
| 84  |                  | <i>Saccharum spontaneum</i> L. KRIB 0077609                                     | Lách                       | GE: 1 (Bleeding)                                                                                                                          | Le                                          | Crushed/ apply to injury places                                               | LC            | B                 |
| 85  | Polygonaceae     | <i>Polygonum aduncum</i> Lour. KRIB 0077750                                     | R sấm, Rau sấm             | DE: 1 (Teeth/gum disease)                                                                                                                 | Ro                                          | Alcohol maceration /chew                                                      |               | B                 |
| 86  | Primulaceae      | <i>Embelia ribes</i> Burm.f. KRIB 0077002                                       | Dương dương                | DE: 1 (Abdominal pain)                                                                                                                    | Wp                                          | Decoction / oral                                                              |               | A                 |
| 87  | Rosaceae         | <i>Rubus alceifolius</i> Poit. KRIB 0077707                                     | Tang R                     | DE: 1 (Liver disease NDS)                                                                                                                 | Le & Ro                                     | Decoction / oral                                                              |               | A                 |
| 88  | Ethnobotanicae   | <i>Conarus pinnatus</i> (Thunb.) Tirveng. KRIB 0077098                          | Rừng, Găng                 | DE: 9 (Teeth/gum disease)                                                                                                                 | St                                          | Rum / dye teeth                                                               |               | B                 |
| 89  |                  | <i>Hedyotis capillaris</i> Wall. ex G. Don KRIB 0077691                         | Pé ngai                    | SE: 1 (Pruritus)                                                                                                                          | Le                                          | Decoction / sauna                                                             |               | B                 |
| 90  |                  | <i>Miconia leucomela</i> Pierre ex P. L. KRIB 0077752                           | Bướm bạc                   | FE: 3 (Fibromyoma uterus, Gonorrhea female, Vaginitis)<br>GE: 4 (Fever/2), Health maintenance/prevention (2)<br>PE: 1 (Sleep disturbance) | St&Le/2), Ro(1)<br>St&Le/2),<br>Wp(2)<br>Le | Decoction / oral                                                              |               |                   |
| 91  |                  | <i>Psychotria indica</i> L. KRIB 0077656                                        | Tè nhũ, Lầu                | DE: 4 (Teeth/gum disease)                                                                                                                 | Ro                                          | Crushed/ chew                                                                 |               | A                 |
| 92  |                  | <i>Wendlandia uvariifolia</i> subsp. <i>laetitia</i> (Pit.) Craun. KRIB 0077746 | Huân lang                  | GE: 2 (Health maintenance)                                                                                                                | Le                                          | Decoction / sauna                                                             |               |                   |
| 93  | Rutaceae         | <i>Citrus × aurantium</i> L. KRIB 0077725                                       | Chanh                      | GE: 6 (Health maintenance/2), Fever/1), Abdominal pain/3)<br>RE: 1 (Influenza)<br>DE: 1 (Teeth/gum disease)                               | Le (3)<br>Ro (5)<br>Le                      | Decoction / sauna (3)<br>Decoction / oral (4)<br>Alcohol maceration/ chew (1) |               | A                 |
| 94  |                  | <i>Citrus maxima</i> (Burm.) Merr. KRIB 0070669                                 | Bưởi, Lá bưởi              | GE: 3 (Health maintenance)<br>PE: 2 (Post-partum symptom)                                                                                 | Le                                          | Decoction / sauna                                                             |               | A                 |
| 95  |                  | <i>Clusia macrocarpa</i> Burm.f. KRIB 0077035                                   | Tè tua                     | DE: 5 (Diarrhoea/2), Teeth/gum disease/2)                                                                                                 | Le & Ro (3)<br>Ro (2)                       | Decoction / oral<br>Crushed/ chew                                             |               | A                 |
| 96  |                  | <i>Zanthoxylum avicennae</i> (Lam.) DC. KRIB 0077601                            | Số số, Sên                 | GE: 2 (Chills)                                                                                                                            | Se                                          | Crushed/ oral                                                                 |               | A                 |
| 97  |                  | <i>Zanthoxylum nitidum</i> (Roxb.) DC. KRIB 0077604                             | Cây trắng                  | GE: 3 (Chills)                                                                                                                            | Se & Le                                     | Crushed/ oral                                                                 |               | A                 |
| 98  | Sapotaceae       | <i>Chrysophyllum cainito</i> L. KRIB 0077627                                    | Vá sữa                     | DE: 1 (Abdominal pain)                                                                                                                    | Le                                          | Decoction / oral                                                              |               | A                 |
| 99  | Scrophulariaceae | <i>Buddleia asiatica</i> Lour. KRIB 0077824                                     | Bọ chó                     | NE: 1 (Headache)<br>CA: 1 (Elevated blood pressure)                                                                                       | St&Le                                       | Decoction / oral                                                              |               | B                 |
| 100 | Sinorabaceae     | <i>Dryocoma longifolia</i> Jack KRIB 0077792                                    | Bách bệnh, Gà pa ca        | SE: 2 (Scabies)<br>DE: 1 (Abdominal pain)                                                                                                 | Le<br>Le & Ro & St                          | Decoction / bath<br>Decoction / oral                                          |               | A                 |
| 101 | Smilacaceae      | <i>Smilax corbularia</i> Rottb. KRIB 0077756                                    | Sà nhũ                     | FE: 3 (Fibromyoma uterus, Gonorrhea female (2))<br>RE: 2 (Spondylitis acute/chronic)                                                      | Ro<br>Ro & St & Le                          | Decoction / oral                                                              |               | B                 |
| 102 |                  | <i>Smilax glabra</i> Rottb. KRIB 0077654                                        | Khúc khúc, Nấm cang        | FE: 3 (Post-partum symptom/complaint oth.)                                                                                                | Ro                                          | Decoction / oral                                                              |               | B                 |
| 103 | Solanaceae       | <i>Capsicum annuum</i> L. KRIB 0018662                                          | Ớt                         | SE: 1 (Burn)                                                                                                                              | Fr                                          | Crushed/ apply to skin                                                        | LC            | B                 |
| 104 |                  | <i>Nicotiana glauca</i> L. KRIB 0077625                                         | Thuốc lá                   | ET: 2 (Injury eye other)<br>GE: 3 (Bleeding)                                                                                              | Le                                          | Rum / fumigation                                                              |               | A                 |
| 105 |                  | <i>Solanum americanum</i> Mill. KRIB 0077773                                    | Sà lầy, Lu la thực         | DE: 1 (Abdominal pain)                                                                                                                    | Le                                          | Decoction / oral                                                              |               | A                 |
| 106 |                  | <i>Solanum torvum</i> Sw. KRIB 0077553                                          | Bà Rạng                    | SE: 1 (Acne)                                                                                                                              | Ro                                          | Decoction / oral                                                              |               | B                 |
| 107 | Theaceae         | <i>Camellia sinensis</i> (L.) Kuntze KRIB 0048067                               | Chè                        | DE: 2 (Liver disease NDS)<br>SE: 2 (Skin disease)                                                                                         | Le                                          | Decoction / oral<br>Decoction / bath                                          |               | A                 |
| 108 | Thymelaeaceae    | <i>Aquilaria crassa</i> Pierre ex Lecomte KRIB 0077645                          | Tầm                        | GE: 1 (Fever)                                                                                                                             | Ro                                          | Decoction / oral                                                              | CR            | A                 |

(continued on next page)

Table 2 (continued)

| No. | Family      | Scientific name, voucher specimen number                 | Van Kien name                 | Uses <sup>1</sup>                                                                | Parts used <sup>2</sup> | Preparation method                                  | IUCN Red list | DVM <sup>3</sup> |
|-----|-------------|----------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------|---------------|------------------|
| 109 | Verbenaceae | <i>Lantana camara</i> L. KRIB 0077778                    | Thục Klay                     | DE: 1 (Abdominal pain)                                                           | Ro                      | Decoction / oral                                    |               | B                |
| 110 |             | <i>Stachytarpheta jamaicensis</i> (L.) Vahl KRIB 0077644 | Đuối chuột, Bật găng, Rà chun | DE: 1 (Abdominal pain)<br>MU: 1 (Low back symptom/complaint)<br>GE: 1 (Bleeding) | Ro<br>Lc                | Decoction / oral<br>Crushed/ apply to injury places |               | A                |
| 111 | Vitaceae    | <i>Cissus quadrangularis</i> L. KRIB 0077810             | Hồ dăng                       | PR: 1 (Post-partum symptom)                                                      | St & Lc                 | Decoction / Sauna                                   |               | A                |

Uses<sup>1</sup>: use-category(# of use-reports: specific uses); GE = General and Unspecified, BL = Blood, Blood Forming Organs and Immune Mechanism, DI = Digestive, EY = Eye, CA = Cardiovascular, MU = Musculoskeletal, NE = Neurological, PS = Psychological, RE = Respiratory, SK = Skin, EN = Endocrine/Metabolic and Nutritional, UR = Urological, PR = Pregnancy, Childbearing, Family Planning, FE = Female Genital.

Parts used<sup>2</sup>: Le = Leaves, St = Stems, Ro = Roots, Fr = Fruits, Wp = Whole plant, La = Latex, Se = Seeds, Ba = Bark.

DVM<sup>3</sup>: Dictionary of medicinal plants in Vietnam, similar use was marked "A" and different use was marked "B".

• • • • • • • • • • •

• • • • • • • • • • •

# Food & Plants as Medicine

Some results of ethnobotanical study on medicinal plants used by local Van Kieu ethnic people of Bac Huong Hoa Nature Reserve, Vietnam (2019):

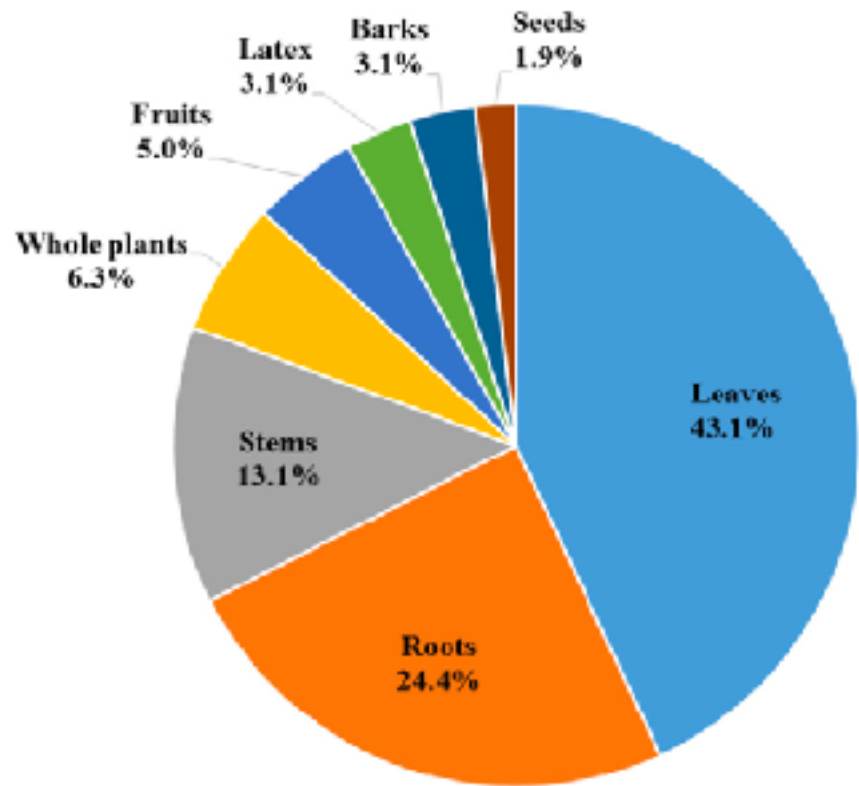


Fig. 3. Plant parts used for their medicinal properties (Percentage).



# Food & Plants as Medicine

Some results of ethnobotanical study on medicinal plants used by local Van Kieu ethnic people of Bac Huong Hoa Nature Reserve, Vietnam (2019):

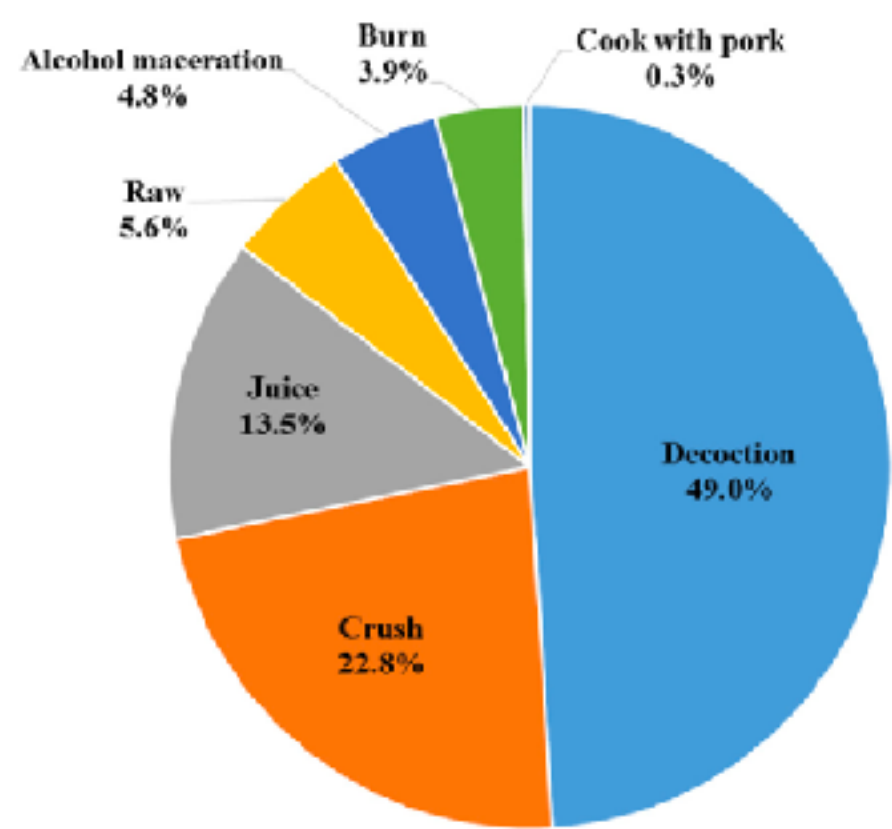


Fig. 4. Modes of preparation used (Percentage).

.....

.....

# Food & Plants as Medicine

Some results of ethnobotanical study on medicinal plants used by local Van Kieu ethnic people of Bac Huong Hoa Nature Reserve, Vietnam (2019):



Bath with red Dao's leaves

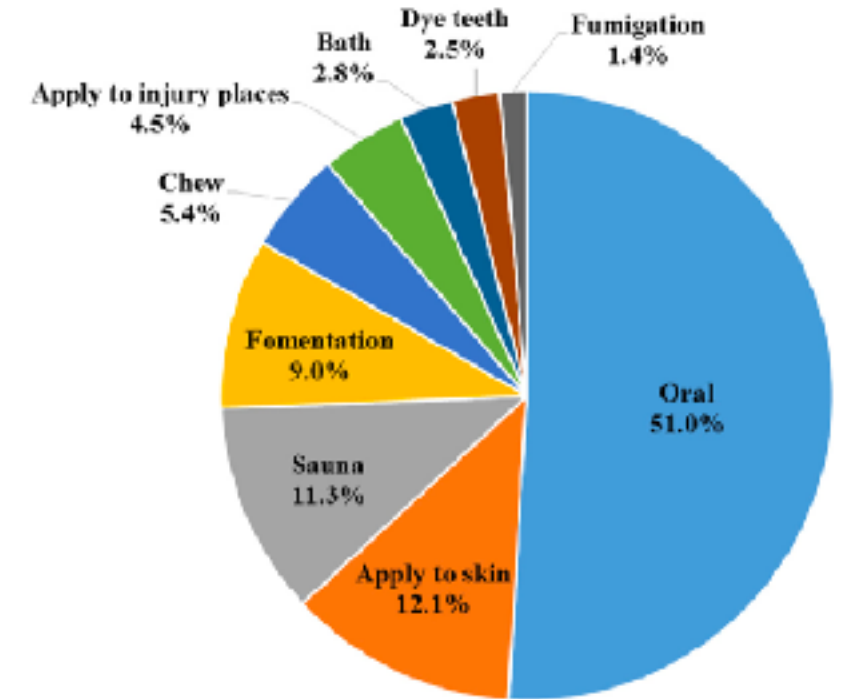


Fig. 5. Modes of administration (Percentage).

# General Health Status of Ethnic Minorities in Vietnam (2024 Study)

Based on a survey of **5,024 ethnic minority individuals** aged 15 and above across 12 provinces:

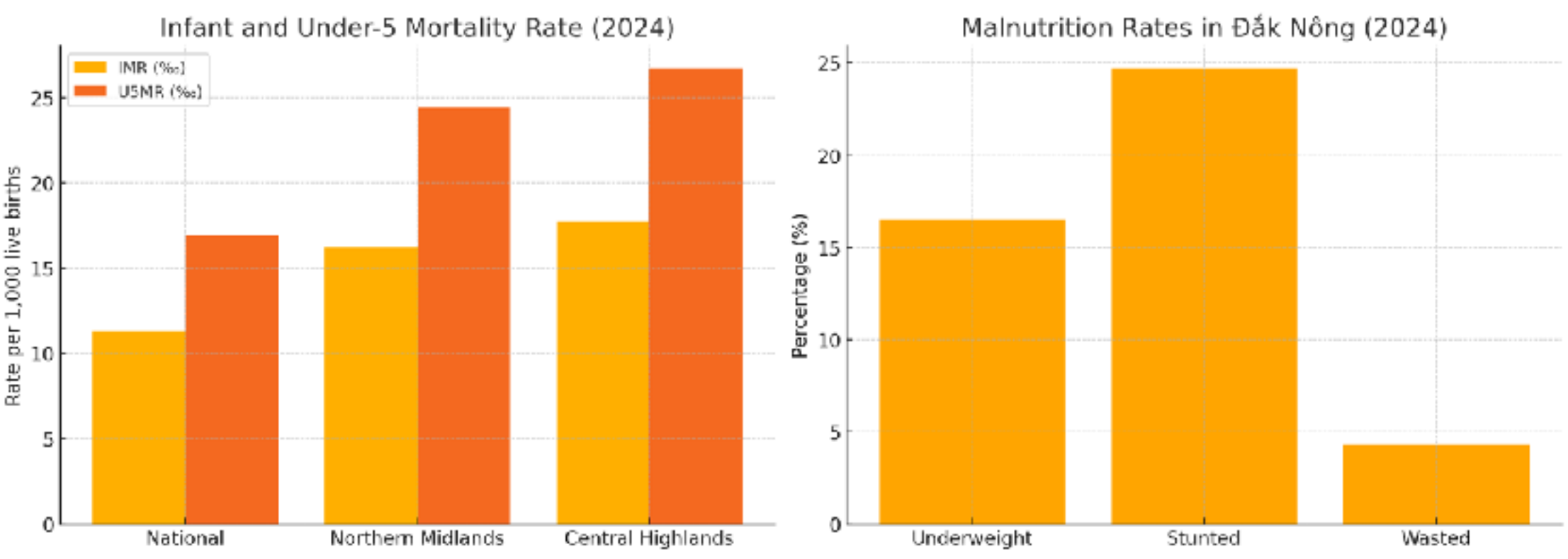
- **73%** reported having been ill in the past three months.
- **57.9%** of elderly respondents were suffering from at least one chronic disease.
- **30.5%** of women aged 15–49 had experienced gynecological conditions.
- **27.4%** of women with children under 5 years old reported that their child had recently suffered from diarrhea.

Vietname Journal of Preventive Medicine, 2024 at URL: <https://vjpm.vn/index.php/vjpm/article/view/1550>

• • • • • • • • • •  
• • • • • • • • • •



# General Health Status of Ethnic Minorities in Vietnam (2024 Study)



Source: <https://sy.t.daknong.gov.vn/giao-duc-suc-khoe/cong-tac-cham-soc-suc-khoe-vung-dong-bao-dan-toc-thieu-so-ngay-cang-duoc-chu-trong.html?>

"How can we improve the health status of ethnic minority communities by integrating modern medicine and healthcare, while still preserving their traditional healing practices?"

• • • • • • • • • •  
• • • • • • • • • •

. . . . .  
. . . . .  
. . . . .

# Indigenous Knowledges

## – Land

## Management

## (Australian context)

. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .

# Fire Farming

Indigenous communities used fire across Australia.

The burning, while reducing risk of uncontrollable fires, also created expansive grasslands, ensured good soils and reproduction of plants which encouraged kangaroos and other animals to come to the regenerated sections, where they could easily be hunted for food.

Selecting what areas to burn, when, with what method and how often, was part of place-specific Indigenous knowledges of the land.

After invasion, fire farming became less used – as the Europeans feared fire.

This meant that thick scrub and tall grasses grew in previously controlled and carefully managed areas, increasing the prevalence of bushfires.



ANU Professor Bill Gammage:

*“Fire can be used for one of three outcomes:*

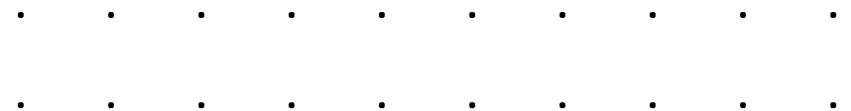
*The first, to encourage native grasses to regenerate and  
produce new feed;*

*the second to reduce scrub and fuel to prevent intense  
bushfires;*

*And, thirdly to promote biodiversity,”*

When laying out burn patters, mob considered not only kangaroos and other hunted animals, but plants, insects and reptiles. The burning patten (or laying out of country), was place and species specific, creating an incredibly complex relationship with the land and ecosystem, which is managed and sustained over multiple generations\*

Plant cycles were especially important – not only for the plants that mob would consume, but the plants stages at which animals would consume them.



Fire knowledge is place and mob specific – what works in one area, may not work in another.

Different methods of fire were used in different times and seasons.

For example, in the NT – we conduct most burns towards the end of the “wet season” / beginning of the dry, when some of the grasses have begun to dry out, but the soil still has some moisture. This ensures the fires will be manageable and productive – and eliminates the risk of out-of-control fires in the dry season.

Across Australia, but mostly in the North, Indigenous mobs are working alongside government and council bodies, farmers and land Management companies to share vital land practice knowledges, including appropriate fire farming techniques.

• • • • • • • • • •  
• • • • • • • • • •



Indigenous Fire Farming Research shows that:

No uncontrolled fires:

There were no uncontrollable fires. Mob understood the risk fire presented to resources, and prevented them through land management practices.

Place-specific templates:

Indigenous people knew their country, knew the species and animals of their country and how to care for them. They new what animals liked to consume what type of plants, and how to ensure these were available, through the use of fire farming.



- • • • • • • • • •
- • • • • • • • • •



. . . . .  
. . . . .  
. . . . .

# Indigenous Knowledges – Land Management (Vietnamese context)

. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .

# Indigenous farming systems

The farming systems reflect deep ecological knowledge, adaptation to harsh mountain environments, and long-term sustainability based on indigenous wisdom.



# Indigenous farming systems

## H'mong (White H'mong – Hà Giang Province)

- **Rock-hole farming (te kho de) (see image):**  
H'mong farmers cultivate maize in limestone rock crevices by building stone borders to retain soil and moisture.
- **Soil improvement techniques:**  
They transport fertile soil and animal manure from valleys, mixing with ash and crop residue to enrich the planting holes.
- **Shifting cultivation:**  
After 4–6 years of use, land is left fallow for natural regeneration before being reused.



• • • • • • • • • •  
• • • • • • • • • •

# Indigenous farming systems

## Tày (Đà Bắc District, Hòa Bình Province)

- **Integrated upland and lowland farming:**  
Tày people combine wet rice cultivation in valleys with upland fields on hillsides for maize, cassava, and other crops.
- **Crop rotation and fallowing:**  
Land is used for 2–3 years with crops like rice and cassava, then left fallow for 2–4 years to restore fertility.
- **VACR model (Vườn - Ao - Chuồng - Rừng / Garden–Fishpond–Livestock–Forest) (see image):**  
A holistic system combining garden crops, aquaculture, animal husbandry, and managed forest plots for sustainability.



. . . . .  
. . . . .

"What challenges do indigenous farming systems face in the modern context, and how can they be addressed?"

• • • • • • • • • •  
• • • • • • • • • •

. . . . .  
. . . . .  
. . . . .

# Indigenous Knowledges – Seasons and Calendars (Vietnamese context)

. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .  
. . . . .



# Indigenous Calendar Systems and Timekeeping

## The Bamboo Calendar of the Mường People (see image)

- The Mường use a traditional timekeeping tool called the “**bamboo calendar**”, made from bamboo segments.
- Each segment is engraved or marked to represent specific days and months of the year.
- This tool helps track time cycles and is used to guide agricultural practices and traditional festivals

## Distinct Ethnic Calendars

- Ethnic groups such as the **H’mông**, **Hà Nhì**, **Cống**, **La Hủ**, **Si La**, **Chăm**, and **Khmer** have **their own lunar-based calendars**, often tied to natural phenomena.
- Examples of traditional New Year celebrations:
  - H’mông: *Tết Nào Pồ Trầu*
  - Hà Nhì: *Tết Hồ Sự Chà*
  - ống: *Tết Ủy La Lóng*
  - La Hủ: *Tết Khộ Xớ*
  - Si La: *Tết Ổ Xị Già*
  - Chăm: *Rija Nugar*
  - Khmer: *Bon Chôl Chnam Thmây*



• • • • • • • • • •  
• • • • • • • • • •



# Seasonal Divisions and Weather Forecasting

## Seasons Based on Agricultural Cycles

- Many ethnic groups divide the year into seasons aligned with farming activities and natural rhythms.
- For instance, the Thái people in Sơn La and Lai Châu celebrate a prolonged “Spring New Year Season” starting with Tết Soong Síp (New Rice Festival), which begins when the rice turns golden ripe.

## Weather Forecasting Through Natural Observation

- Groups like the Thái and H’mông predict weather by closely observing plants and animal behavior.
- Example: Changes in palm tree leaves or animal movements may signal an approaching storm or winter season.

• • • • • • • • • •  
• • • • • • • • • •

# Festivals Aligned with Seasonal and Agricultural Cycles

Traditional festivals of Vietnam's ethnic minorities are closely tied to seasonal rhythms and farming activities.

Examples:

- Thái people with Xên bản, Xên Mường festival, held Around the 2nd lunar month to honor ancestors and pray for peace and prosperity in the community.
- Giáy people (Tả Van, Sa Pa – Lào Cai) with Roóng Poọc Festival, Held on Dragon Day in the first lunar month to pray for a bountiful harvest and favorable weather.



# Week 5

## What will be covered in week 5:

- Astronomy from the Indigenous context
- Case study: Bamboo trees
- Case study: Lotus farming
- Water Sciences

